C12 # Agent SDK & IDE Integrations

Design Advanced

Integrating Claude Code into Xcode, VS Code, and other environments

Two products, separate surfaces to verify

Claude Code is an interactive development product. **Claude Agent SDK** supports integrations or applications that call Claude. Their exact relationship depends on the version and surface published by Anthropic; do not infer a shared architecture from a similar interface.

Surface	Check before deciding
IDE extension	Editor documentation and official product page
SDK	API reference, supported versions, and permission model
MCP	Server, exposed tools, transport, and requested permissions

MCP and IDE integration

MCP is a tooling protocol. An integration can use MCP, an SDK, a native extension, or several of these mechanisms. None should be presented as a required condition for another.

```
IDE or local service ↔ published integration ↔ authorized tools
```

The security decision is unchanged: identify the tools actually exposed, their scope, and the execution path before granting permission.

MCP vetting before installation

```
# 1. Check the maintainer, repository, and targeted release
gh repo view <mcp-repo>

# 2. Check the version (never use "latest")
# Always pin: @1.2.3 not @latest

# 3. Read tools and permissions, then scan if available
# Follow the current scanner installation and scope documentation
# https://github.com/snyk/agent-scan
```

Do not use stars or a ranking as security evidence. Never approve an MCP from an unknown source without checking its tools, permissions, version, and update mechanism.

Custom Agent SDK use cases

The Agent SDK can support integrations beyond standard IDEs: CI, PR analysis, dashboards, or review systems. Check the current API and limits in official documentation before designing the integration. See also Agent Tools and the glossary.